

# Ciarán Flanagan

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## EDUCATION

### Northeastern University

Honors Candidate for B.S. in Computer Science & Biology

May 2027

GPA: 3.8

- **Awards:** Honors Program, Dean's List (5x)
- **Involvement:** Galway Boston GAA (Gaelic Football, Hurling), Climbing Club, Run Club, Intramurals (Basketball, Soccer)
- **Relevant Coursework:** Software Engineering, Algorithms & Data, Theory of Computation, Object-Oriented Design, Databases

## TECHNICAL SKILLS

**Languages:** Java, Python, Go, SQL, TypeScript, JavaScript, Kotlin, Swift, Bash, Racket

**Frameworks/Tools:** React, FastAPI, Express.js, Material-UI, Pydantic, Zod, TanStack Query, Pandas, NumPy, Scikit-learn, Claude Code

**DevOps & Databases:** Docker, AWS, PostgreSQL, MySQL, MongoDB, SQLite, Git, GitHub Actions, CI/CD, Terraform, Prometheus

## EXPERIENCE

### Walmart Global Tech, Bentonville, AR

Jun 2026 – Aug 2026

Software Engineer Intern

- Cut new-market CI/CD onboarding from 2 days to 15 minutes by extending the Kotlin and Swift generators that scaffold new market apps across Walmart's iOS and Android monorepos, supporting growth from 11 to 14 international markets
- Architected the extension as composable scaffold-only and CI-only CLI modes with fully idempotent runs, letting engineers provision an app, its pipeline, or both, and safely re-run without duplicating files or clobbering existing configuration
- Extended an internal MCP server and orchestration layer to thread CI configuration across four services end-to-end, passing credentials in Walmart's encrypted secret-reference format and backfilling only placeholder values, never overwriting real config
- Shipped a stale PR bot with 30-day inactivity and 7-day auto-close thresholds, replacing a GitHub action unavailable on internal infrastructure, and closed 2,000+ abandoned PRs across both monorepos in its first month of production use

### NExT Consulting (Client: Via Separations), Boston, MA

Jul 2025 – Dec 2025

Software Engineer Co-op

- Built a full-stack warehouse management system with React/TypeScript, FastAPI/Python, and PostgreSQL, architecting a 50+ table relational schema to model end-to-end inventory operations and reviewing code for a 6-person engineering team
- Shipped a two-phase inventory move workflow and lot-splitting system with parent-child item tracking, integrating location scanning and generating audit trails so any unit's movement history stayed traceable across every warehouse location
- Architected a three-tier exception handling system with domain-specific exceptions and global handlers, decoupling business logic from HTTP transport and rolling back transactions automatically to prevent partial database writes on failure
- Eliminated N+1 query patterns by enforcing explicit relationship loading at the ORM layer, and protected data integrity with partial unique constraints for soft-deleted records and validation blocking deletes on rows still referenced elsewhere
- Contributed to the system's AWS cloud infrastructure design and implemented role-based access control in custom middleware, then built a 7-workflow GitHub Actions pipeline running automated tests and build validation on every pull request

### Northeastern University, Boston, MA

Aug 2024 – Present

Teaching Assistant, Software Engineering & Fundamentals of Computer Science I/II

- Taught test-driven development, design patterns, and agile practices in TypeScript, functional programming and recursion in Racket, and object-oriented design in Java, spanning the intro sequence through upper-level software engineering coursework
- Led weekly labs and office hours, evaluating code quality and testing strategy on student submissions while guiding implementations of recursive algorithms, immutable data structures, interfaces, and object-oriented class hierarchy design

### OASIS (Student Organization), Boston, MA

May 2024 – May 2026

Executive Board President

- Led a 120-student cohort and matched mentors to 25 teams building full-stack web applications, writing curriculum on React, Node.js, PostgreSQL, and API design, delivering 8 technical workshops, coordinating with Khoury clubs, and maintaining the org website

## PROJECTS

### bioflow (Go, Docker, Prometheus, GitHub Actions)

- Built a DAG-based job execution engine in Go that runs containerized pipeline steps across a bounded worker pool, resolving step dependencies into concurrent waves with cycle detection so independent steps execute in parallel without manual ordering
- Implemented content-addressed caching and crash-safe checkpointing so interrupted runs resume from the last completed step and unchanged steps are skipped, with per-step retries, exponential backoff, and Prometheus instrumentation

### agenteval (Python, SQLite, pytest, GitHub Actions)

- Built a Python tracing and evaluation library for LLM agents that captures tool calls, token counts, latency, and cost per run, then diffs two recorded versions to flag metric regressions and surface newly failing cases separately from aggregate metrics